

Notes: Bigio & La'O (QJE, 2020)

Distortions in Production Networks

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1 Big picture

- **Question.** How does an input-output network transmit sectoral distortions to macro aggregates? In particular, when a tax, markup, or financing friction hits a sector, does it show up mainly in aggregate productivity or in the aggregate labor wedge?
- **Framework.** The paper studies a static multisector economy with intermediate inputs, monopolistic competition within sectors, generic constant-returns technologies, and elastic household labor supply.
- **Central result.** Near efficiency, sectoral distortions have **zero first-order effects on TFP** but **nonzero first-order effects on the labor wedge**. The sufficient statistic for those first-order labor-wedge effects is the vector of **Domar weights**:

$$\frac{\partial \log A}{\partial \log \phi_i} = 0, \quad \frac{\partial \log \Lambda}{\partial \log \phi_i} = \lambda_i.$$

- **Why this is a big deal.** Hulten’s theorem says that, in efficient economies, first-order TFP effects of productivity shocks are Domar-weighted. Bigio–La’O show the parallel statement for *distortions*: the first-order macro object is not TFP, but the labor wedge.
- **Cobb-Douglas specialization.** In a Cobb-Douglas input-output economy, those first-order effects can be written in terms of the Leontief inverse / network centrality. Distortions in more central sectors move the labor wedge more.
- **Bottleneck lesson.** A distortion in a bottleneck sector can have a *large* effect on the labor wedge while doing almost *nothing* to TFP. The reason is that it wedges many price paths without generating much misallocation.
- **Quantitative application.** The paper calibrates the network to U.S. BEA input-output tables and maps Gilchrist–Zakrajsek credit spreads into within-period financing wedges during the 2008–09 financial crisis.
- **Headline quantitative result.** The U.S. input-output network amplified financial distortions by about a factor of **two**. In the calibration, the labor wedge falls much more than TFP because distortions act on the labor wedge at first order but on TFP only at second order near efficiency.
- **Deep intuition.** Productivity shocks *change resources*. Distortions *change prices and allocations*. Near efficiency, the reallocation inefficiency shows up only quadratically in TFP but linearly in the aggregate markup / labor wedge.

2 Data and measurement (theoretical objects and empirical inputs)

2.1 Theoretical objects that organize the paper

- **Primitive state.** The original state is

$$s = (z, \tau, \theta, \delta),$$

where z are sectoral productivities, τ revenue taxes or subsidies, θ sectoral elasticities of substitution, and δ the fraction of tax proceeds that is effectively “thrown out.”

- **Reduced-form distortions.** The paper rewrites the economy in terms of:

$$\phi_i(s) \equiv \frac{\theta_i(s) - 1}{\theta_i(s)} \frac{1 - \tau_i(s)}{1 - \delta_i(s)\tau_i(s)}, \quad \psi_i(s) \equiv 1 - \delta_i(s)\tau_i(s), \quad (1)$$

and net output

$$q_i(s) \equiv \psi_i(s)y_i(s) = \psi_i(s)z_i(s)F_i(\ell_i(s), x_i(s)). \quad (2)$$

- **Interpretation.**

- ϕ_i is the **true reduced-form distortion**: it wedges marginal rates of transformation against marginal rates of substitution.
- ψ_i is a **wasted distortion**: it acts like a productivity shock because some output is literally discarded.

- **Net productivity.** Since wasted distortions are isomorphic to productivities, the paper defines

$$a_i(s) \equiv \psi_i(s)z_i(s)$$

and studies the reduced state $s = (a, \phi)$.

- **Aggregate wedges.** The prototype economy is summarized by:

$$C(s) = A(s)L(s), \quad \frac{V'(L(s))}{U'(C(s))} = \Lambda(s) \frac{dC(s)}{dL(s)}.$$

Here $A(s)$ is the efficiency wedge / TFP and $\Lambda(s)$ is the labor wedge.

- **Domar weights.** At the efficient steady state,

$$\lambda_i \equiv \frac{p_i y_i}{C}.$$

These sales shares are the sufficient statistics that drive the paper's main first-order results.

2.2 Empirical inputs for the quantitative application

- **Input-output data.** The authors use annual BEA input-output tables, beginning in 1997, at the three-digit NAICS level.
- **Observed sectoral objects.** For each sector i and year t , they observe total sales $p_{it}y_{it}$, intermediate expenditures $p_{jt}x_{ij,t}$, labor expenditures $W_t\ell_{it}$, and final use $p_{it}c_{it}$.
- **Calibrated Cobb-Douglas shares.** These data pin down:

$$\alpha_{it} = \frac{W_t\ell_{it}}{W_t\ell_{it} + \sum_j p_{jt}x_{ij,t}}, \quad v_{it} = \frac{p_{it}c_{it}}{\sum_i p_{it}c_{it}}, \quad g_{ij,t} = \frac{p_{jt}x_{ij,t}}{\sum_j p_{jt}x_{ij,t}}. \quad (3)$$

The paper then uses the 2007 values of these shares in the crisis calibration.

- **Financial-distortion proxy.** Sectoral wedges come from the Gilchrist–Zakrajsek excess bond premium, denoted r_{it}^{GZ} .

- **Mapping spreads to wedges.** In the working-capital interpretation,

$$\phi_{it} = \frac{1}{1 + r_{it}^{GZ}}, \quad (4)$$

after abstracting from markups.

- **Returns to scale / fixed-factor correction.** To relax pure CRS, the paper uses:

$$y_{it} = z_{it} \left(\ell_{it}^{\alpha_i} x_{it}^{1-\alpha_i} \right)^{\eta_i}, \quad (5)$$

where $\eta_i \in (0, 1]$ is either interpreted as decreasing returns or as the income share of labor plus intermediates.

- **Key identifying ratio.** The expenditure-to-sales ratio is

$$\varepsilon_{it} \equiv \frac{W_t \ell_{it} + \sum_j p_{jt} x_{ij,t}}{p_{it} y_{it}}. \quad (6)$$

Under the model, $\varepsilon_{it} = \phi_{it} \eta_i$.

- **Calibration range for returns to scale.** The paper sets

$$\bar{\eta}_i \equiv \max_t \varepsilon_{it}, \quad \eta_i \in [\bar{\eta}_i, 1].$$

The lower endpoint gives the “lowest possible DRS” model; the upper endpoint gives the CRS benchmark.

3 Models and empirical specifications (all equations + interpretation)

3.1 Environment: firms, intermediate goods, and household

Firm k in sector i produces

$$y_{i,k} = z_i F_i(\ell_{i,k}, x_{i,k}), \quad (7)$$

where the composite intermediate input is

$$x_{i,k} \equiv G_i(x_{i,k}), \quad (8)$$

and $x_{i,k} = (x_{i1,k}, \dots, x_{iN,k})'$ collects purchases of sectoral goods.

Within each sector, differentiated outputs are aggregated into a uniform sectoral good through

$$y_i = \left(\int y_{i,k}^{\frac{\theta_i-1}{\theta_i}} dk \right)^{\frac{\theta_i}{\theta_i-1}}. \quad (9)$$

The representative household has preferences

$$U(C) - V(L), \quad (10)$$

with final consumption aggregator

$$C = C(c_1, \dots, c_N). \quad (11)$$

Goods-market clearing is

$$y_i = c_i + h_i + \sum_{j \in I} x_{ji}, \quad (12)$$

and labor-market clearing is

$$L = \sum_{i \in I} \ell_i. \quad (13)$$

- **Interpretation.** This is a static Long–Plosser style production network, except that production choices may be distorted and household labor is elastic, which creates room for an aggregate labor wedge.
- **Key conceptual distinction.** Productivities z_i shift the feasible production frontier. Distortions wedge firms' private decisions away from that frontier.

3.2 Equilibrium characterization in reduced form

Proposition 1 says equilibrium allocations are characterized by the following two first-order conditions:

$$V'(L(s)) = \phi_i(s) U'(C(s)) \frac{\partial C(s)}{\partial c_i(s)} \frac{\partial q_i(s)}{\partial \ell_i(s)}, \quad (14)$$

$$\frac{\partial C(s)/\partial c_j(s)}{\partial C(s)/\partial c_i(s)} = \phi_i(s) \frac{\partial q_i(s)}{\partial x_{ij}(s)} \frac{\partial G_i(s)}{\partial x_{ij}(s)}, \quad (15)$$

together with net-output resource constraints

$$q_i(s) = c_i(s) + \sum_{j \in I} x_{ji}(s). \quad (16)$$

- **Interpretation of the first condition.** The household's marginal disutility of labor is equated to the social marginal value of sector- i labor *times* the distortion ϕ_i .
- **Interpretation of the second condition.** Relative marginal utilities across consumption goods are equated to relative marginal products across sectors, again up to the same sectoral wedge.
- **Why the change of variables matters.** Replacing gross output y_i by net output $q_i = \psi_i y_i$ separates two forces:
 - **true distortions** ϕ_i , which generate labor wedges and misallocation;
 - **wasted distortions** ψ_i , which are observationally equivalent to productivities.

3.3 Financial frictions as a special distortion

With working-capital frictions, firms must finance wages and intermediate inputs at an intraperiod interest rate $1 + r_i$. In that case Proposition 2 shows:

$$\phi_i(s) = \frac{\theta_i(s) - 1}{\theta_i(s)} \frac{1}{1 + r_i(s)}, \quad \psi_i(s) = 1. \quad (17)$$

- **Interpretation.** Higher short-run borrowing costs distort production exactly like a tax / markup wedge.
- **Why this matters for the application.** It lets the authors map observed credit spreads directly into sectoral distortions without building a full dynamic collateral model.

3.4 Prototype economy and aggregate wedges

The multisector equilibrium is mapped into the benchmark wedge-accounting prototype:

$$C(s) = A(s)L(s), \quad (18)$$

$$\frac{V'(L(s))}{U'(C(s))} = \Lambda(s) \frac{dC(s)}{dL(s)}. \quad (19)$$

- **Efficiency wedge $A(s)$.** This is aggregate TFP: the shift in the aggregate production relation between labor and final consumption.
- **Labor wedge $\Lambda(s)$.** This is the gap between the marginal rate of substitution and the marginal rate of transformation. It is also interpretable as the aggregate markup or a tax-like distortion on labor/consumption.
- **Big organizing idea.** The entire paper asks which sectoral shocks move A and which move Λ .

3.5 Hulten for productivity shocks and Hulten-like theorem for distortions

At the efficient steady state, Domar weights are

$$\lambda_i = \frac{p_i y_i}{C}. \quad (20)$$

The first benchmark result is Hulten's theorem:

$$\frac{\partial \log A}{\partial \log a_i} = \lambda_i. \quad (21)$$

The paper's main contribution is the parallel result for distortions:

$$\frac{\partial \log A}{\partial \log \phi_i} = 0, \quad \frac{\partial \log \Lambda}{\partial \log \phi_i} = \lambda_i. \quad (22)$$

- **Economic meaning.** Near efficiency, a small distortion does not lower TFP at first order because the efficient allocation is already a local maximum of productive efficiency.
- **But the same distortion does move the labor wedge at first order.** Distorted sectoral prices compound along supply chains, so the final good's price deviates from marginal cost even when TFP barely moves.
- **Most important takeaway.** Productivity shocks are first-order *TFP* shocks; distortion shocks are first-order *labor-wedge* shocks.

3.6 Cobb-Douglas special case: Leontief inverse and network centrality

For the tractable special case, technologies and preferences are

$$F_i(\ell_i, x_i) = \ell_i^{\alpha_i} x_i^{1-\alpha_i}, \quad G_i(x_i) = \prod_{j \in I} x_{ij}^{g_{ij}}, \quad C(c_1, \dots, c_N) = \prod_{i \in I} c_i^{v_i}. \quad (23)$$

Let G be the input-output matrix and define the Leontief inverse

$$\mathbf{L} \equiv (I_N - \text{diag}(1 - \alpha)G)^{-1}. \quad (24)$$

Lemma 1 shows that relative prices satisfy

$$\log p(s) - \log W(s) = -\mathbf{L}(\log a(s) + \log \phi(s)) + \kappa_p. \quad (25)$$

The influence vector is

$$\beta' \equiv v' \mathbf{L}. \quad (26)$$

Theorem 3 then gives

$$\frac{\partial \log A}{\partial \log a_i} = \beta_i, \quad (27)$$

and near efficiency,

$$\frac{\partial \log \Lambda}{\partial \log \phi_i} = \beta_i. \quad (28)$$

- **Interpretation.** $\mathbf{L}$ sums all direct and indirect downstream propagation effects.
- **Network-centrality result.** Distortions in sectors with high downstream centrality have larger aggregate labor-wedge effects.
- **Why price propagation matters.** Productivity and distortion shocks have similar *relative-price* effects, but not the same *allocation* effects. Productivity changes real resources; distortions mainly change how those resources are priced and allocated.

3.7 Supply chains, pure vertical limits, and bottlenecks

To build intuition away from small-shock approximations, the paper studies a supply-chain economy. In the pure vertical limit, the economy collapses to a single labor-to-consumption path and Proposition 3 implies

$$A(s) = 1, \quad \Lambda(s) = \prod_{i \in I} \phi_i(s), \quad \chi_i(s) \equiv \frac{\partial \log \Lambda}{\partial \log \phi_i}(s) = 1. \quad (29)$$

- **Interpretation.** In a pure vertical chain there is no scope for misallocation, so TFP does not move. But every wedge passes one-for-one into the labor wedge.

In the general Cobb-Douglas economy, the paper defines supply- and demand-bottleneck measures:

$$b_i^S(s) \equiv \alpha' \left[\text{diag}(\phi(s))^{-1} - G' \text{diag}(1 - \alpha) \right]^{-1} e_i, \quad (30)$$

$$b_i^D(s) \equiv v' \left[\text{diag}(\phi(s))^{-1} - G' \text{diag}(1 - \alpha) \right]^{-1} e_i. \quad (31)$$

Proposition 5 gives the elasticity of the labor wedge to a sectoral distortion:

$$\chi_i(s) \equiv \frac{\partial \log \Lambda}{\partial \log \phi_i}(s) = \frac{1}{\phi_i(s)} \Lambda(s)^{-1} b_i^S(s) b_i^D(s), \quad (32)$$

and the corresponding TFP elasticity:

$$\frac{\partial \log A}{\partial \log \phi_i}(s) = \beta_i - \chi_i(s). \quad (33)$$

- **Economic meaning.** A bottleneck distortion raises many output prices because many labor-to-consumption paths pass through that sector.
- **Core insight.** Bottleneck sectors can be extremely powerful for the labor wedge even when they generate little measured TFP loss.

3.8 Quantitative multiplier and back-of-the-envelope formula

The authors compare the calibrated U.S. input-output economy to an otherwise equivalent *horizontal* economy with no intermediate linkages. With small symmetric distortions, the CRS network economy satisfies

$$d \log \Lambda^{IO} = \left(\sum_{i \in I} \lambda_i \right) d \log \bar{\phi}, \quad (34)$$

while the horizontal economy satisfies

$$d \log \Lambda^{hor} = d \log \bar{\phi}. \quad (35)$$

Therefore the labor-wedge network multiplier is approximately

$$\frac{d \log \Lambda^{IO}}{d \log \Lambda^{hor}} = \sum_{i \in I} \lambda_i = \frac{\sum_i p_i y_i}{C}. \quad (36)$$

- **Interpretation.** The multiplier is approximately total sales over GDP.
- **Quantitative implication.** In the U.S. data this ratio is about 2, which is exactly why the calibrated network multiplier lands near 2.
- **Connection to roundabout economies.** In a one-sector roundabout economy this collapses to the familiar $1/\alpha$ multiplier from Basu and Jones.

4 Identification, implementation, and what makes the application credible

4.1 What makes the theory credible

- **No IV or natural experiment.** This is not a causal reduced-form paper. Credibility comes from equilibrium accounting plus exact first-order theory.
- **Key identification object.** Theorems 1 and 2 pin down the first-order mapping from micro shocks to aggregate wedges. That is the theoretical source of identification.

- **Why the result is sharp.** Once one accepts the environment, the distinction between productivity shocks and distortion shocks is exact to first order:
 - productivity shocks load on TFP;
 - distortion shocks load on the labor wedge.

4.2 Why the quantitative comparison is informative

- **Equivalent-horizontal benchmark.** The paper compares the U.S. input-output economy to an otherwise equivalent economy without intermediate linkages. That benchmark cleanly isolates the role of the *network*.
- **Symmetric vs asymmetric distortions.** Symmetric distortions identify the amplification created by network structure alone. Asymmetric distortions then show what happens with the actual cross-sectional pattern in the GZ data.
- **Use of GZ spreads.** The Gilchrist–Zakrajsek premium is designed to net out default risk, so the paper interprets it as a time-varying cost of external finance rather than simply expected default losses.
- **Cross-check with expenditure shares.** The paper shows that sectoral changes in the GZ-based wedge co-move positively with changes in the expenditure-to-sales ratio, which is exactly what the model predicts if financing costs distort input use.

4.3 Implementation issues and limitations emphasized by the paper

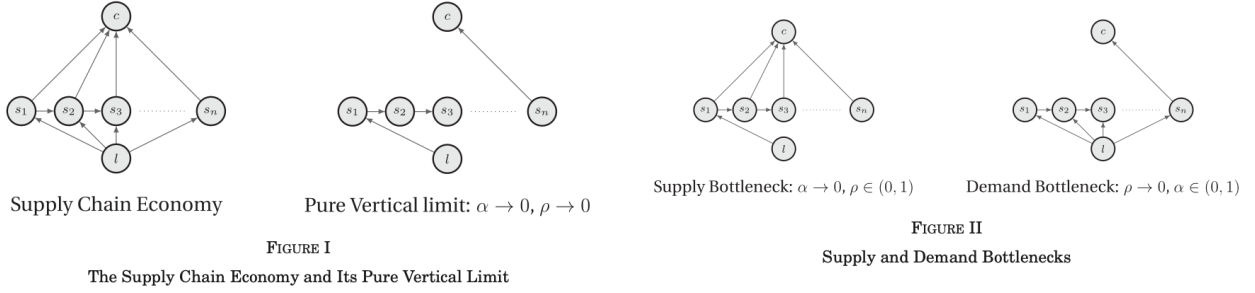
- **Static environment.** The model abstracts from capital accumulation and standard collateral dynamics. It is closer to a within-period working-capital story than to a full financial accelerator.
- **Elasticity restrictions.** The quantitative application uses Cobb-Douglas technology, so substitution elasticities are pinned to one.
- **Perfect labor mobility.** Labor can move freely across sectors in the calibration. Slower reallocation would likely strengthen the distortion effects.
- **Sector-level wedge measurement.** The GZ series is observed at the sector level, not at the individual-firm link level. So the application is best interpreted as a disciplined illustration, not a full structural estimate of all credit frictions in the economy.
- **DRS ambiguity.** Because $\varepsilon_{it} = \phi_{it}\eta_i$, the paper cannot separately identify wedges and decreasing returns. It therefore reports a range between the lowest DRS model and the CRS benchmark.

5 Tables and figures

5.1 Figure I: The supply chain economy and its pure vertical limit

Read:

- The figure shows a chain with multiple labor-to-consumption paths on the left and the pure vertical limit on the right.
- The visual lesson is that when all paths collapse into one, there is no room for misallocation. Distortions then show up purely as labor-wedge effects, not as TFP effects.
- This figure is the paper’s cleanest intuition for why TFP can be insensitive to distortions even when aggregate markups move a lot.



5.2 Figure II: Supply and demand bottlenecks

Read:

- The left panel isolates a **supply bottleneck**: all labor must first pass through the upstream sector.
- The right panel isolates a **demand bottleneck**: all output must eventually be funneled through the downstream sector before reaching final consumption.
- The figure teaches the paper’s strongest nonlocal result: some sectors matter disproportionately because many production paths run through them.

5.3 Figure III: Cross-sectional means and standard deviations

Read:

- The left panel plots the cross-sectional mean of $\Delta \log \phi_{it}$ against the mean of $\Delta \log \varepsilon_{it}$; they comove closely, especially around the crisis.
- The right panel plots the cross-sectional standard deviations of those same variables and shows greater dispersion in recession periods.
- The empirical message is that the GZ-based distortion proxy is not arbitrary: it lines up with the expenditure-to-sales object the model says should move with wedges.

5.4 Table I: The labor wedge network multiplier during the 2008–09 financial crisis

Read:

- This is the headline quantitative table.

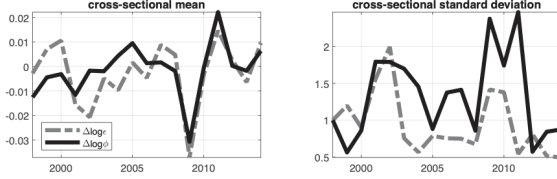


FIGURE III
Cross-Sectional Means and Standard Deviations

TABLE I
THE LABOR WEDGE NETWORK MULTIPLIER DURING THE 2008–09 FINANCIAL CRISIS

Returns to scale	$\Delta \log A$		$\Delta \log \Lambda$	
	DRS	CRS	DRS	CRS
Horizontal, symmetric ϕ	0	0	-0.0307	-0.0307
U.S. IO network, symmetric ϕ	-0.0012	-0.0027	-0.0523	-0.0650
U.S. IO network, asymmetric ϕ	-0.0013	-0.0028	-0.0493	-0.0591
Network multiplier	∞	∞	1.70	2.12

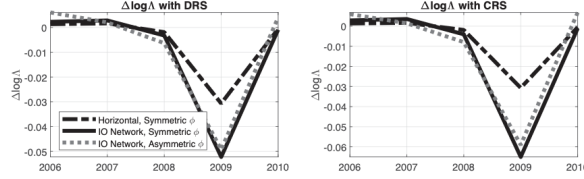


FIGURE IV
The Model-Generated Labor Wedge

The model-generated labor wedge for the horizontal economy with symmetric shocks (dark dashed), the calibrated network economy with symmetric shocks (dark solid), and the calibrated network economy with asymmetric shocks (gray dotted) for 2006–10. Left: lowest DRS model. Right: CRS model.

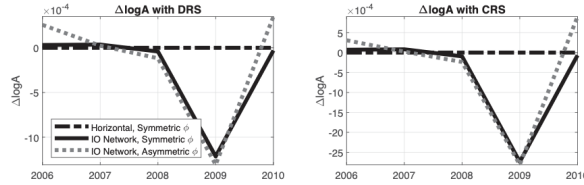


FIGURE V
The Model-Generated Efficiency Wedge

The model-generated efficiency wedge for the horizontal economy with symmetric shocks (dark dashed), the calibrated network economy with symmetric shocks (dark solid), and the calibrated network economy with asymmetric shocks (gray dotted) for 2006–10. Left: lowest DRS model. Right: CRS model.

- In the horizontal benchmark, symmetric distortions lower the labor wedge by about 3.07% and leave TFP unchanged.
- In the calibrated U.S. network, symmetric distortions lower the labor wedge by about 5.23% in the DRS model and 6.50% in the CRS model.
- The implied labor-wedge network multiplier is therefore about 1.70 under lowest DRS and 2.12 under CRS.
- With asymmetric GZ distortions, the labor wedge still falls much more than TFP: about 4.93% vs 0.13% in DRS, and 5.91% vs 0.28% in CRS.

5.5 Figure IV: The model-generated labor wedge

Read:

- The horizontal economy with symmetric shocks produces a much smaller fall in the labor wedge than the U.S. network economy.
- The gap between the dashed horizontal-economy line and the solid network-economy line is the visual measure of network amplification.

- The dotted asymmetric-shock line sits close to the solid line in crisis years, showing that the empirical cross-sectional pattern does not overturn the main multiplier story.

5.6 Figure V: The model-generated efficiency wedge

Read:

- TFP barely moves in all cases relative to the labor wedge.
- In the horizontal economy with symmetric distortions, TFP is flat by construction because symmetric wedges do not generate misallocation.
- Even in the full network model, the TFP movement is tiny compared with the labor-wedge movement. This directly visualizes the paper’s main theorem.

6 Short summary

- **Conclusion.** Sectoral distortions in production networks show up mainly as first-order movements in the aggregate labor wedge, not as first-order TFP losses.
- **Intuition.** Distortions wedge prices away from marginal costs all along supply chains. Near efficiency, that compounded pricing effect is linear for the labor wedge but only quadratic for TFP.
- **Empirical lesson.** In the U.S. 2008–09 crisis, the input-output network roughly doubled the macro effect of financing distortions and most of that amplification appeared in the labor wedge.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main theoretical result.** There is a Hulten-like theorem for sectoral distortions:

$$\frac{\partial \log \Lambda}{\partial \log \phi_i} = \lambda_i$$

near efficiency, while

$$\frac{\partial \log A}{\partial \log \phi_i} = 0.$$

- **Structural implication.** In Cobb-Douglas economies, those effects are governed by the Leontief inverse / influence vector, so network centrality determines aggregate importance.
- **Bottleneck implication.** Distortions in bottleneck sectors are especially powerful for the labor wedge because they distort many price paths at once.
- **Quantitative implication.** In the crisis calibration, the U.S. network amplified financing distortions by about a factor of two.
- **Measurement implication.** If an economist sees large labor-wedge movement with little TFP movement, the theory points toward distortion shocks rather than productivity shocks.

7.2 Economic intuition

- Productivity shocks create or destroy productive capacity, so they move TFP directly.
- Distortions mostly reshuffle choices inside a fixed technology set. Near the efficient allocation, those reallocations are locally second order for productive efficiency.
- But the same distortions alter sectoral prices immediately, and those relative-price wedges compound through supplier-customer chains. That is why the aggregate markup / labor wedge moves at first order.
- The input-output network matters because it decides *how many times* and *through which sectors* the distortion is transmitted before reaching final demand.

7.3 Takeaway

This paper gives a clean micro-to-macro map for distortions in production networks. Its punchline is not just that networks amplify shocks, but that *the object they amplify depends on the shock*: productivity shocks load on TFP, while distortion shocks load on the labor wedge. The crisis calibration makes that distinction concrete by showing that financial distortions in 2008–09 were strongly amplified by the U.S. production network and that most of the action appeared in the aggregate markup rather than in measured TFP.